Response to Reviewer 1

Manuscript: *Proxy-Based Diagnostics of Quantum Oracle Sketching Robustness for Non-IID Sensor and Telemetry Streams*
Journal: *Sensors* (MDPI) · **Manuscript ID:** sensors-4470240

Dear Editor,

Thank you for the opportunity to submit a revised version of our manuscript, *Proxy-Based Diagnostics of Quantum Oracle Sketching Robustness for Non-IID Sensor and Telemetry Streams* (Manuscript ID: sensors-4470240), to *Sensors*. We appreciate the time and effort the reviewer has dedicated to the paper, and we are grateful for the constructive comments, which have improved it considerably.

We agree with the central point of the review. The previous version, although hedged internally, still read as evidence about quantum performance, whereas what the work can honestly support is a proxy-based diagnostic and hypothesis-generation framework for correlation-sensitive streaming analysis. The revision repositions the manuscript accordingly, in the title, abstract, statistics, figures and conclusions, without removing any experiment or altering any measured result. All stochastic outputs regenerate identically from the same seeded pipeline, which we verified before beginning the re-analysis. In accordance with the Academic Editor's request that preprints not be cited, the Zhao et al. (2026) preprint has been removed from the manuscript and the reference list, which now contains 54 references; no other work was substituted for it, and the affected statements were revised so that each remaining reference supports only what it actually establishes.

The changes are marked in the highlighted version submitted with this letter. A point-by-point response to the comments follows.

Comments from Reviewer 1

Comment 1: *I strongly recommend to reduce the emphasis on quantum advantage and instead presenting the τ,r landscape primarily as a diagnostic framework to identify scenarios where future quantum implementations is most promising.*

Response: Thank you for this recommendation, which we have adopted as the organising principle of the revision. The manuscript is now presented as a diagnostic and hypothesis-generation framework rather than as evidence of quantum advantage. The title now leads with "Proxy-Based Diagnostics"; the Introduction states that the paper "neither demonstrates nor claims a realised quantum-classical separation"; and the Conclusions identify the (tau, r) landscape as the primary contribution, with hypothesis generation as the secondary one. Section 7.4 states three falsifiable hypotheses, H1 to H3, each with an explicit refutation criterion, so that the question of where future implementations are most promising becomes concrete and testable. Throughout the paper, "quantum-favourable" has been replaced by "proxy-favourable" or "hypothesized favourable".

Comment 2: *Please clarify the relationship between the heuristic proxy model and the original theoretical framework.*

Response: Thank you for raising this. A dedicated subsection now draws that boundary explicitly. Section 3.3 separates five things: the published quantum-streaming results that provide the general context, together with the statement that these are problem-specific and do not establish the operational tau, r, T_eff or accuracy model used here; the assumptions this paper adopts; the heuristic constructions it introduces, namely the effective-sample correction in T_eff and the VC-type task rate; the memory curves, retained only as illustrative scaling references; and the scope limitations. Table 1 records the provenance and epistemic status of every model-level quantity, with a column stating what is not theoretically guaranteed. The relationship is therefore presented as one of motivation and analogy rather than derivation, and the manuscript states that it adds no quantum theorem.

Comment 3: *The analytic Markov τ proxy differs substantially from the empirical measurements. I recommend to discuss this discrepancy in greater detail and explain whether it reflects limitations of the proxy model or the synthetic data generation process.*

Response: We agree that this deserved fuller treatment, and Section 7.3 is now devoted to it. For Markov switching the two quantities answer different questions: each bit's marginal autocorrelation decays as rho^k and crosses 1/e at about 2.8 lags at rho = 0.7, whereas the joint n-bit chain requires on the order of n log n / (1 - rho) steps to mix. Both values are correct answers to different questions, so the divergence is partly expected behaviour and partly a limitation of summarising dependence in a single scalar. It is not an artefact of the data generation: the generators reproduce their intended statistics, and the gap persists under all four alternative tau estimators examined in Experiment 9, which isolates the estimator family as the cause. The subsection also sets out the practical consequences, namely that the empirical tau serves as the operational input, the analytic tau as a conservative envelope, and their ratio as a diagnostic for long-memory content.

Comment 4: *I recommend to validate the proposed framework on additional real-world datasets. It will help to better demonstrate its generalizability.*

Response: We agree that this is the most valuable extension, and we are explicit that it has not been carried out in the present revision. Adding datasets would have required a new experimental campaign, and we preferred to keep this revision re-analysis based so that every published number remains reproducible from the pinned seeds. What we have added, from the same raw data, is an encoder and resolution sensitivity analysis in Section 6.8: the two streams were re-encoded across ten combinations of sampling resolution and bucket width, and the ordering of NYC Taxi above Machine Temperature holds in all ten. The reported placement is therefore not an artefact of one encoder setting. The manuscript now states that the real-stream experiment is illustrative rather than statistically representative, the Threats section records the two-dataset limitation, and Future Work item (iii) commits to network telemetry, cybersecurity event streams, financial series and IoT fleets, with hypothesis H1 defining the protocol by which generalisability will be tested.

Comment 5: *Please include a more comprehensive sensitivity analysis of the selected proxy parameters, particularly C and δ, by this you will show how they influence the reported results.*

Response: Agree, and we have added this quantitatively. Because the proxy is a closed-form function of the measured tau and r, it can be re-evaluated on the same per-seed values across C in {1, 1.5, 2, 2.5} crossed with delta in {0.01, 0.05, 0.1}. The new Table 8 reports the resulting movement of the proxy-baseline crossover. The ordering of the regimes is invariant, since C and delta enter only through the monotone factor C^2 log(1/delta), but the crossover location moves from about rho* = 0.38 at C = 2.5 and delta = 0.01 to beyond the sweep range at C = 1, with rho* about 0.82 at the operating point. The text now instructs that the landscape be read ordinally, the abstract carries this conditionality, and Section 5.3 states that C = 2 is a conventional order-unity choice not fixed by any constant in the cited literature.

Comment 6: *The comparison between deterministic quantum proxies and stochastic classical baselines should be presented more cautiously. The current statistical analysis may unintentionally suggest a comparison between implemented algorithms rather than between empirical results and theoretical proxies.*

Response: This is an important point and we have rebuilt the statistical presentation around it. Table 4 now reports Hodges-Lehmann estimates of the paired difference with exact signed-rank 95% intervals and matched-pairs rank-biserial correlations, and the p-values are retained only as descriptive companions. A dedicated paragraph in Section 6.5 explains that the pairing is between a stochastic learner and a model prediction, that the proxy's seed-to-seed variation comes solely from the per-seed tau estimate, and that the rank-biserial value here indexes sign concordance across seeds rather than magnitude. The same paragraph states that the tests "do not, and cannot, establish that one algorithm outperforms another, and no quantum algorithm is implemented anywhere in this comparison". We also disclose that none of the five comparisons survives Holm correction at n = 10, that the interval at rho = 0.48 is not robust to a single step of the integer-valued tau estimator, and that the crossover location is a deterministic solution whose uncertainty derives from the estimator and the constants rather than from the tests.

Comment 7: *I recommend to provide a stronger justification for the selected classical baselines and discussing if additional streaming methods can offer a more informative comparison.*

Response: Thank you for this suggestion. Section 5.2 now justifies the choice explicitly. The three baselines span complementary axes of bounded-memory streaming learning, namely gradient-based (online SGD), variance-reduced gradient-based (averaged SGD) and non-gradient hash-count-based (Count-Min), while sharing the single-pass, fixed-memory regime that the proxy models. We also explain why drift-adaptive methods such as ADWIN-equipped learners and adaptive random forests are deliberately excluded from the primary comparison: they respond to non-stationarity by discarding history, which would confound the correlation-budget question the proxy isolates. Benchmarking them on the same landscape is named as an extension in Future Work.

Comment 8: *Please explain more clearly why the proposed operational proxies, τ and r, are sufficient to characterize the correlation properties of the studied data streams, and discuss any limitations of relying only on these two measures.*

Response: We agree, and the revision now treats two-scalar sufficiency as an assumption rather than a fact. Section 3.3 lists it under Assumptions as a modelling choice of this paper, and Section 7.3 shows its failure mode: for non-mixing, long-range-dependent processes there is no single scalar on which the analytic and empirical estimators could agree, so a single-number tau is intrinsically lossy in that regime. Two mitigations are stated. The ratio of analytic to empirical tau is itself a useful third diagnostic that flags long-memory content, and multi-scale refinements of tau, using integrated-autocorrelation estimates at several horizons, are named as follow-on work. The Threats section records the residual risk under construct validity.

Comment 9: *I recommend to expand the discussion of the assumptions behind each synthetic correlation regime and to please explain how closely these settings represent practical sensor and telemetry environments.*

Response: Thank you for this suggestion. Section 4 already carried per-regime provenance notes, and the summary in Section 4.6 now adds the realism mapping. Markov switching idealises short-range state persistence in sensor readouts; seasonal drift represents the diurnal and weekly cycles of telemetry and demand series; burst repetition represents the duplicate-heavy event bursts of monitoring and logging pipelines; and long-range dependence represents the self-similar behaviour first measured on Ethernet traffic. We state plainly that these are idealisations, that real streams mix the mechanisms, as the two NAB cases show, and that the mapping to any concrete deployment is qualitative.

Comment 10: *The manuscript would benefit from a dedicated discussion of threats to validity, include the assumptions introduced by the heuristic proxy model, synthetic data generation and feature binarization.*

Response: Agree. This is now a dedicated section. Section 8, "Threats to Validity", is organised by construct, internal, external and statistical-conclusion validity, and covers the items you list. It addresses the heuristic proxy constructs, with pointers to the provenance table; the selection of synthetic generators; feature binarisation, noting that arcsine shrinkage makes the real-stream tau and r estimates lower bounds on the correlation of the underlying continuous process; the design of the deterministic-versus-stochastic comparison; dataset selection; the definition of the target, including the full-series threshold look-ahead raised in review; the fixed hyperparameters; and the ten-seed budget.

Comment 11: *I recommend to discuss the computational complexity of estimating τ and r and the scalability of the proposed framework for high-volume streaming systems.*

Response: Thank you for pointing this out. Section 5.1 now contains a paragraph on estimator cost and scalability. The tau estimator requires the bitwise autocorrelation up to lag K, which costs O(nTK) time and O(nK) memory using a ring buffer per feature. The r estimator scans a forward window of length W, costing O(TW) comparisons, or expected O(T) with hashing, and O(Wn) memory. Both are linear in T, independent of the alphabet size, single-pass friendly and parallel across features. On the longest stream studied, T = 22,695, they complete in well under a second, so the diagnostic adds negligible overhead in high-volume deployments.

Comment 12: *Some figures contain many overlapping curves and annotations, I recommend to simplify the visual presentation and enlarge labels to improve readability.*

Response: Thank you for this observation. All nine figures were re-audited and several were regenerated. The legend entry “Proxy-model advantage” in Figure 5 has been renamed “Proxy-estimated separation”, and stray typographic characters were removed from five panel titles. In-image labels that referred to qubit counts and to quantum order-notation were replaced with neutral proxy labels in Figures 1, 2 and 6. The synthetic markers in Figure 8(d) are now identified as schematic orientation marks, and the caption gives the numeric stream coordinates instead of relying on visual adjacency. Confidence-interval statements in captions were scoped to the stochastic panels only, and four tables that previously overran the text width were reformatted. All regenerations are rendering changes only; the underlying numerical outputs are unchanged.

Comment 13: *The discussion could better distinguish theoretical quantum memory advantages from the practical observations obtained through the proxy-based evaluation.*

Response: We agree, and this distinction is now enforced wherever the memory curves appear. Experiment 6 opens with the statement that no memory lower bound is measured empirically in that experiment and that no result constitutes an empirical memory floor for online SGD, averaged SGD or Count-Min. Section 3.3, under “Memory references”, records that the published quantum-streaming papers do not establish the O(n) and Omega(sqrt(N)) pair for the diagnostic task studied here, and that the pair is retained solely as a paper-defined illustrative scaling reference. Table 1 lists that pair as “introduced here (illustrative scaling reference)”, with “a proven separation for this diagnostic task” in the not-guaranteed column. The caption of Figure 6 labels panel (a) as an illustrative memory-scaling reference with no measured content, and Section 7.1 closes its three structural factors with the statement that all three are properties of the proxy model or its illustrative references and that none is a measured quantum result. The statement about n >= 20 is marked as an extrapolation beyond the tested range.

Comment 14: *Please discuss the applicability of the proposed framework to other streaming domains beyond sensor telemetry, such as network traffic, financial streams, or cybersecurity monitoring.*

Response: Thank you for this suggestion. Table 9 now maps network backbone traffic, financial returns, sensor and IoT streams, and concept-drifted streams onto the landscape using published measurement ranges, with the values recomputed from the paper’s own formula, the verdicts given as ranges, and the estimator convention stated. Future Work item (iii) commits to the domains you name: network telemetry in the form of flow and latency series, cybersecurity event streams such as intrusion and log-anomaly feeds, financial tick and returns series, and large-scale IoT monitoring fleets.

Comment 15: *strengthen the conclusion, for this emphasize the practical contributions of the study, clearly summarize its limitations, and please outline the most important directions for future research.*

Response: Agree. The Conclusions have been rewritten to this structure. A paragraph on the practical contribution presents the landscape as a low-cost pre-screening step: estimate tau and r, place the stream, and read off whether correlation alone erodes the effective budget below the level needed for rare-event learning, as it does for the Machine Temperature stream at T_eff = 52. A second paragraph on limitations and required validation points to the Threats section and names the decisive missing step, namely an implemented or simulated sketching pipeline evaluated at coordinates the landscape predicts to be favourable and unfavourable. The research-question answers are restated in effect-size terms with the estimator convention made explicit in each case.

We hope that the revised manuscript now meets the standards of *Sensors*, and we look forward to hearing from you in due course. We would be glad to respond to any further questions or comments.

Sincerely,

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